Repository : A repository is a storage location where a project's files and their version history are kept using Git**.**

Types of GitHub Repositories

1. Public Repository

- Anyone on GitHub can view the repository.
- Anyone can clone or download the code.
- Best for open-source projects and portfolios.

2. Private Repository

- Only you and the people you give access to can view the repository.
- Used for personal, college, or confidential projects.

Basic Git Commands :

1. **git init :** Initializes a Git repository in a folder.
   It allows Git to start tracking your project files**.**

```
C:\Users\NAMRATA\OneDrive\Desktop\Github> git init
Reinitialized existing Git repository in C:/Users/NAMRATA/OneDrive/Desktop/Github/.git/
```

2. **git status :** Shows the current status of your files.
   It tells you which files are changed, added, or ready to commit.

```
C:\Users\NAMRATA\OneDrive\Desktop\Github> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

3. **git add . :** Adds all changed files to the staging area.
   It prepares the files for a commit.

```
C:\Users\NAMRATA\OneDrive\Desktop\Github> git add .
```

4. **git config –list :** Displays all Git configuration settings.
   It shows your username, email, and other Git settings.

```
C:\Users\NAMRATA\OneDrive\Desktop\Github> git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Namrata Chavanke
user.email=namratachavanke9@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/namratachavanke9/VSEC_Assignment.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main
```

5. **git config user.email :** Shows the email address set in Git.
   It displays the email used for your commits.

```
C:\Users\NAMRATA\OneDrive\Desktop\Github> git config user.email "namratachavanke9@gmail.com"

C:\Users\NAMRATA\OneDrive\Desktop\Github>
```

6. **git remote add origin <repository-url> :** Connects your local repository to a GitHub repository. It adds the remote named origin.

```
C:\Users\NAMRATA\OneDrive\Desktop\Github>git remote add origin https://github.com/namratachavanke9/VSEC_Assignment.git
```

7. **git commit -m "your message"** : Saves the staged changes with a message.
   It creates a new commit in your local repository.

```
C:\Users\NAMRATA\OneDrive\Desktop\Github> git commit -m "Added HTML and JavaScript assignments"
[master (root-commit) 205981c] Added HTML and JavaScript assignments
 4 files changed, 430 insertions(+)
 create mode 100644 Assignment_1.html
 create mode 100644 Assignment_3.html
 create mode 100644 Assignment_3.js
 create mode 100644 Assignment_4.js
```

8. **git branch -M main :** Renames the current branch to main.
   It is commonly used before pushing to GitHub.

```
C:\Users\NAMRATA\OneDrive\Desktop\Github> git branch -M main
```

9. **git push -u origin main :** Uploads the main branch to GitHub for the first time.
   It also links the local branch with the remote branch.

```
C:\Users\NAMRATA\OneDrive\Desktop\Github>git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 3.24 KiB | 3.24 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/namratachavanke9/VSEC_Assignment.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```